

Harshana Lakshara Fernando

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SUMMARY

Full-Stack Software Engineer experienced in end-to-end application delivery, legacy integrations, and production support. Strong in **Java**, **Spring Boot**, **Kafka**, **React**, and DevOps workflows. Known for rapid learning, taking ownership of distributed systems, and choosing solutions that balance reliability, scalability, and real product needs.

EXPERIENCE

R&D Software Engineer — KōreroCare Platform Mar 2026 – Present
Empathic Computing Laboratory, University of Auckland Remote

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern — ISA Aviation Technology Dec 2024 – May 2025
Information Systems Associates (Pvt) Ltd Colombo, Sri Lanka

- Owned design and delivery of a distributed **Spring Boot/Kafka**-backed PNR ID generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
- Designed a **pool-based PNR allocation model** to replace legacy Oracle-sequence generation, supporting short domain-specific PNR formats while avoiding predictable sequential identifiers.
- Implemented **concurrency-safe** PNR allocation using a **Kafka-backed pool**, ensuring the same PNR was not assigned to multiple service instances, while using Java **ReentrantLock** for application-level critical sections.
- Led the rollout from legacy PNR generation to AeroOrder using database-driven **feature toggles** and a **strangler-pattern** migration approach for phased production cutover and fallback.
- Resolved production **Kafka consumer lag** and offset-reset issues by adding graceful shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
- Integrated **gRPC** clients and Kafka consumers into the legacy **Java 8** AeroMart monolith for backward-compatible communication with AeroOrder microservices.
- Built async load-testing and monitoring support using **JMeter**, **Prometheus**, and **Grafana** to validate PNR/order flows, queue time, consumer behavior, and service latency.
- Added Jenkins environment support, maintained **>90% unit test coverage**, and coordinated delivery with QA and distributed engineering teams.

Freelance Software Engineer — Selected Client Projects May 2025 – May 2026
Client Platforms Remote / Sri Lanka

- Delivered Agrinova, a full-stack multi-branch inventory platform (Vercel, Supabase/PostgreSQL), implementing transaction-safe stock logic across 3 core workflows (issuance, returns, purchases).
- Reduced repeat dashboard data-fetch latency from ~384ms to <1ms for cached views using React Query caching and pagination.
- Owned end-to-end client-facing fixes, resolving production authentication, version compatibility, and workflow-level validation bugs.

PROJECTS

Employee Management System [Architecture] Java 17, Spring Boot, Spring Cloud, Microservices, Kubernetes
Built cloud-native microservices using Eureka, Gateway, RabbitMQ-based config refresh, and Resilience4j (**Circuit Breaker**, **Rate Limiter**).

GoTogether - Intelligent Travel Platform [Architecture] Next.js, Go, Spring Boot, Keycloak, Gemini API, PostgreSQL
Developed a full-stack **Next.js** and **Spring Boot** platform featuring **AI-driven** itinerary generation via Gemini API and Keycloak OAuth2.

Event-Driven Food Delivery Platform Java, Spring Boot, React.js, Kafka, Redis, WebSockets, PostgreSQL
Built an event-driven full-stack app combining a **React.js** UI with a **Spring Boot** backend, utilizing **Kafka** and WebSockets.

Serverless Auction Platform [Architecture] Node.js, AWS Lambda, API Gateway, DynamoDB, SQS/SNS
Architected event-driven auction/auth services using **AWS Lambda**, EventBridge, **DynamoDB** GSIs, and custom JWT authorizers.

RPAL Interpreter Python, pytest, TDD, Data Structures
Developed an interpreter for the RPAL programming language implementing Lexical/Syntax analysis, AST standardization, and a CSE machine using strict **Test-Driven Development** (TDD) with **pytest**.

More projects, architecture diagrams, and articles are available on my portfolio.

SKILLS

Backend: **Java**, **Spring Boot**, Go/Gin, Python, Node.js, REST, **gRPC**, GraphQL, Hibernate

Frontend: **React**, Next.js, **TypeScript**, JavaScript, React Query

Messaging & APIs: **Kafka**, RabbitMQ, WebSockets, Protobuf, JWT, OAuth2/OIDC

Data: **PostgreSQL**, MySQL, Oracle DB, Redis, Couchbase, DynamoDB, Prisma

Cloud / DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux

Observability / Testing: Prometheus, Grafana, JMeter, Gatling, JUnit, Mockito, Pytest

EDUCATION

B.Sc. (Hons) in Engineering — Computer Science and Engineering, University of Moratuwa 2022 – 2026
G.C.E. A/L: 3 A's, Physical Science Stream, District Rank 17

TRAINING & CERTIFICATIONS

CKA Training (KodeKloud, 2025) · AWS SAA-C03 Training (Udemy, 2025) · Apache Kafka Series (I, II, III) · 100 Days of DevOps (KodeKloud, 2025) · Advanced OAuth Security (Udemy, 2024)